

# A Secure and Low-Latency Design for LDACS Ground Station Handover

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**Abstract**—L-band Digital Aeronautical Communications System (LDACS) has been developed to provide a secure and high-throughput aviation communication service. Unfortunately, the state-of-the-art of the LDACS handover mechanism fails to protect station identity information and incurs excessive handover latency. In order to tackle these issues, this paper proposes a secure and low-latency handover scheme, with the aid of anonymous identity and of aircrafts' trajectories. The handover scheme proposed is comprised of two stages, namely, pre-handover and handover. In contrast to the current handover scheme where the ground station is selected for the maximum received power, in our pre-handover stage, the ground station to be connected is predictively pre-selected following the aircraft's trajectory planned. As a benefit, the authentication between the aircraft and the ground station to be connected can be prepared before handover. To realize this, a periodic renewal of temporary identity code is introduced for each key establishment, to attain the anonymity both of aircraft and ground station. Following this, the token including the aforementioned encryption authentication result is sent to the aircraft. In this case, the complicated mutual authentication and key establishment between the aircraft and the ground station to be connected can be avoided, which may accelerate the handover process. As for the handover stage, the aircraft's anonymous identity information along with the token acquired are sent to the next ground station. Then, the aircraft's authenticated result in the token is verified using the private key of LDACS access network. As such, the aircraft can be quickly handed over to the ground station to be connected. To validate the scheme proposed, we rigorously evaluate its security performance using Tamarin Prover and quantify the handover latency relying on OMNeT++ 6.0. It is verified that the scheme proposed is capable of effectively safeguarding station identity information during the handover process, while simultaneously reducing the handover latency.

**Index Terms**—LDACS, anonymity, low-latency, handover.

## I. INTRODUCTION

Secure and efficient aeronautical communication is the cornerstone of air traffic management [1]. However, with the growth of civil aviation traffic and the increase in digital services, traditional aeronautical communication systems, such as VHF Datalink mode 2 (VDLm2), which rely on dedicated and limited spectrum resources, have reached their efficiency and data rate limits. This often leads to issues such as frequency congestion, posing significant challenges to the secure and efficient operation of Air Traffic Services (ATS).

This work was supported by the National Key Research and Development Program of China under Grant 2023YFB3907001.

To address the future demands for digitalization, high bandwidth, and enhanced security in aeronautical communication systems, the L-band Digital Aeronautical Communications System (LDACS) [2], suitable for En-Route air-to-ground communication, has been proposed. LDACS utilizes the L-band spectrum (960–1164 MHz) and offers advantages such as high spectrum utilization efficiency, large bandwidth, and support for high data transmission rates. During the LDACS communication, aircraft along flight routes need to handover between different ground stations as they traverse different regions to maintain continuous connectivity [3].

From the outset of its standardization, LDACS has given full consideration to the design of secure handover mechanism. However, constructing a secure handover protocol for LDACS is no simple task. First, the design of the LDACS secure handover protocol must meet specific security requirements, ensuring resilience against common attacks and interference. Second, the data involved in the secure handover process should not consume excessive spectrum resources or introduce significant delays; efforts should be made to minimize data overhead and reduce latency. To address these challenges, Nils et al. made significant contributions to the design of secure handover mechanisms for LDACS. In 2018, they proposed the LDACS security architecture along with corresponding security algorithms, which enable LDACS to achieve security objectives within this framework [4]. In 2022, Nils and colleagues introduced a fundamentally secure handover procedure for LDACS, fulfilling security and performance requirements for both data and voice communication over LDACS [5]. In 2024, Khan et al. employed post-quantum cryptography techniques to protect the LDACS secure handover process from potential post-quantum threats [6].

Furthermore, anonymity has also become a prominent research focus in the study of handover authentication. In 2024, Yang et al. have proposed a secure communication protocol for space information networks featuring periodic k-times anonymous authentication. This protocol enables satellite users to perform up to k anonymous authentications with ground stations within a specified time period [7]. As for trajectory-based pre-handover authentication methods closely related to this paper, in 2020, Ruhui et al. proposed a fixed trajectory pre-handover authentication scheme for mobile relay nodes in high-speed rail networks [8].

The goal of this article is to propose a secure and low-latency handover scheme for LDACS, leveraging anonymous identity protection to safeguard station identity information, and utilizing pre-handover based on aircraft trajectories to reduce the actual handover latency. The contributions and novelty of our study are summarized as follows:

- We safeguard station identity privacy by employing temporary identity code. The anonymous transmission and periodic update of station identity information enhance the protection of station data and provide robust defense against interception by external attackers.
- We divide the LDACS handover process into two stages: pre-handover and handover. In the pre-handover stage, the next ground station to be accessed is determined based on the aircraft's trajectory, and key establishment with aircraft is completed in advance.
- In the handover stage, the aircraft seamlessly accesses the ground station to be connected using the authentication token issued by the departing ground station, thereby significantly reducing both the latency and air-ground message overhead during the handover process.

This paper is organized as follows: Section II provides an overview of the technical background of LDACS. Section III details the proposed handover protocol, while Section IV presents a comprehensive security analysis of the protocol. Section V evaluates the protocol's security features and performance through comparative analysis. Finally, Section VI concludes the paper with a summary of key findings.

## II. BACKGROUND ON LDACS

LDACS is a digital bidirectional communication system based on Orthogonal Frequency Division Multiplexing (OFDM) [9], designed for secure and efficient communication between aircraft station (AS) and ground station (GS). It meets the requirements for ATS, Aeronautical Operational Control (AOC), and foreseeable future communication scenarios, and is currently being standardized by International Civil Aviation Organization (ICAO) [10] and Internet Engineering Task Force (IETF) [11]. The LDACS system architecture connects to the Aeronautical Telecommunication Network (ATN) via the ground access network. Data is transmitted between the LDACS ground stations and the ATN through the access network router (AC-R), with transport layer security provided by Datagram Transport Layer Security (DTLS) and network layer security ensured by Internet Protocol Security (IPSec). When data is transmitted between the LDACS GS and the AS, the security of the communication is protected by the LDACS security mechanisms.

Each LDACS GS can simultaneously support up to 512 AS. The GS transmits data to the AS via the Forward Link (FL), while the AS transmits data to the GS via the Reverse Link (RL). In the FL, the GS first announces the presence of the LDACS ground cell through Broadcast Channel (BCCH), then allocates resources to the AS via the Common Control Channel (CCCH), and finally transmits user data through the Data Channel (DCH). In the RL, the AS station uses the resources

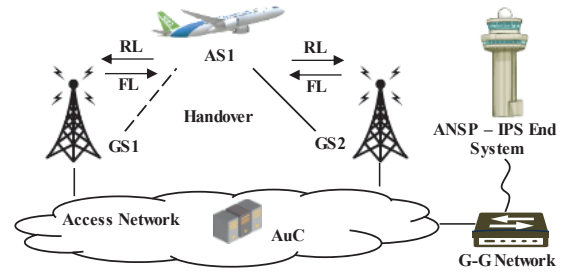


Fig. 1. Security design framework of LDACS. AS refers to Aircraft Station, GS refers to Ground Station, AuC refers to Authentication Center, RL refers to Reverse Link, FL refers to Forward Link, G-G Network refers to Ground-Ground Network, ANSP-IPS End System refers to Air Navigation Service Provider-Internet Protocol Suite End System.

allocated by the GS to send data in short bursts. Access to the LDACS cell is requested through Random Access Channel (RACH), while permission to transmit user data is requested via the Dedicated Control Channel (DC), and user data is subsequently transmitted via the DCH [12].

The LDACS protocol stack consists of the Physical Layer (PHY), Medium Access Layer (MAC), Data Link Service Layer (DLS), Link Management Entity (LME), Voice Interface (VI), and Sub-Network Protocol Layer (SNP) [13]. The PHY is responsible for the transmission of data over the radio channel. The MAC is tasked with organizing and managing the transmission within time slots and frequency. The DLS provides different Classes of Service (CoS), depending on the priority of the user packet, which are mapped into the DLS accordingly. The LME primarily handles the configuration, resource management, and mobility management of LDACS. The SNP, functioning as a network layer connector, interconnects the LDACS network and provides end-to-end user and control plane connectivity between the AS and GS. Therefore, the design of the LDACS secure handover mechanism must be closely integrated with the LDACS protocol stack.

In the design of the LDACS security mechanism, Public Key Infrastructure (PKI) serves as the foundation for station identity authentication. The LDACS PKI architecture typically consists of an Advisory Council that oversees certificate policies (CP), a Management Authority that manages the PKI in accordance with these policies and authorizes the issuance of sub-Certificate Authorities (CA), with the root CA being responsible for the sub-CA, and the sub-CA being responsible for the identity certificates of the AS and GS. LDACS digital certificates are based on X.509 version 3 certificates. Within the LDACS security mechanism, the sub-CA is typically handled by the Authentication Center (AuC) of the ground access network. Therefore, when considering the design of the LDACS secure handover, the overall framework consists of the AS, GS, and AuC, as shown in Fig. 1.

To maintain the security of LDACS, the LDACS security specification establishes eight objectives that the designed security mechanisms must meet: (1) Ensuring system availability and continuity; (2) Protecting message integrity; (3)

Ensuring message authenticity; (4) Providing non-repudiation of message origin; (5) Safeguarding message confidentiality; (6) Offering authentication functions; (7) Authorizing allowed operations for users while denying unauthorized actions; and (8) Preventing intrusions within the LDACS domain and their propagation beyond the domain.

### III. PROPOSED HANDOVER PROTOCOL

To protect the anonymity of identity information during LDACS handover and reduce handover latency, this section proposes an LDACS handover protocol based on temporary identity code and AS's trajectory. As defined by the official LDACS specification [14], the focus of this proposed handover protocol is on the first case, specifically handover type 2. That is, the handover between two ground stations within the same ANSP LDAN (LDACS Access Network).

Our design follows the principles outlined below: (1) During signaling interactions, a Temporary Unique Addresses (TUA) code [15] is used to replace the real Unique Addresses (UA), ensuring station anonymity and preventing UA code leakage. (2) The security algorithms at the terminal are simplified as much as possible to conserve computational resources. (3) The interaction between AS and GS during handover is minimized to save spectrum resources and reduce latency. This protocol enables LDACS aircraft handover between ground stations, with the specific protocol flow depicted in Fig. 2.

Otherwise, in the operation of the LDACS system, several assumptions are made: (1) The AS and GS entities are susceptible to attacks. (2) The AuC is fully trusted. (3) The AuC has issued trusted certificates, UA, and TUA for both aircraft and ground platforms. (4) The AS has sent trajectory information to GS1 and established a secure connection. (5) A secure connection has also been established between GS1 and GS2. (6) This protocol currently considers only the pre-quantum phase. Table I provides the symbols and terminology used in the protocol. The overall protocol is divided into two phases: pre-handover and handover. The detailed protocol flow is explained as follows.

#### A. Pre-Handover

- **Step 1: GS1 informs GS2 of the essential information regarding the AS.** After GS1 and the AS complete mutual authentication and key establishment (MAKE), GS1 searches for the next ground station (GS2) that the AS will reach based on its trajectory information. GS1 measures the current timestamp  $T_{GS1}$  and concatenates  $P_{AS}$ ,  $k_{voice,RL}$ ,  $k_{BC}$  to generate  $k_{set}$ . It then computes  $m_{GS1} = T_{GS1} || TUA_{GS2} || SAC_{GS2} || k_{set} || TUA_{AS} || SAC_{AS} || algo$ . Subsequently, GS1 generates a random number  $N_1$  and encrypts  $m_{GS1}$  using the Authenticated Encryption with Associated Data (AEAD) encryption algorithm with the DCH session key  $k_{GS1,GS2}$  shared between GS1 and GS2. This results in  $(c_{m_{GS1}}, t_{m_{GS1}}) \leftarrow AEAD_{Encrypt}(k_{GS1,GS2}, N_1, m_{GS1})$ . GS1 then sends  $N_1$ ,  $c_{m_{GS1}}$  and  $t_{m_{GS1}}$  to GS2.

TABLE I  
NOTATION FOR PROTOCOL

| Field          | Description                     |
|----------------|---------------------------------|
| $UA_A$         | Unique Address of A             |
| $TUA_A$        | Temporary Unique Address of A   |
| $SAC_A$        | Sub-net Access Code of A        |
| $k_A$          | Key of A                        |
| $list$         | Cipher-Suite Options for LDACS  |
| $algo$         | Cipher-Suite choice from $list$ |
| $Trajectory_A$ | Trajectory of A                 |
| $T_A$          | Timestamp of A                  |
| $P_A$          | Public part of DHKE of A        |
| $x_A$          | Private part of DHKE of A       |
| $m_A$          | Message of A                    |
| $N$            | Nonce                           |
| $c_A$          | Ciphertext of A                 |
| $t_A$          | Tag of A                        |
| $PubKey_A$     | Public Key of A                 |
| $PrivKey_A$    | Private Key of A                |

- **Step 2: GS2 sends its key information and the newly allocated SAC for the AS to GS1.** GS2 receives  $N_1$ ,  $c_{m_{GS1}}$  and  $t_{m_{GS1}}$  and decrypts the message using the shared DCH session key  $k_{GS1,GS2}$ :  $m_{GS1} \leftarrow AEAD_{Decrypt}(k_{GS1,GS2}, N_1, c_{m_{GS1}}, t_{m_{GS1}})$ . First, GS2 verifies the timestamp  $T_{GS1}$ . If it falls within the acceptable time threshold, the process continues; otherwise, the process is immediately terminated. GS2 then stores  $P_{AS}$ ,  $k_{voice,RL}$ ,  $k_{BC}$ ,  $TUA_{AS}$ ,  $SAC_{AS}$ , and  $algo$ . GS2 generates its public-private key pair  $(P_{GS2}, x_{GS2}) \leftarrow DHToken(\cdot)$  using Elliptic Curve Diffie-Hellman Ephemeral (ECDHE) and computes the shared key  $z \leftarrow DHSharedSecret(P_{AS}, x_{GS2})$ . The shared key  $z$  is then used to derive the session key  $k_{AS,GS2} \leftarrow HKDF(z)$  using the HKDF key derivation algorithm. GS2 measures the current timestamp  $T_{GS2}$  and assigns a new Sub-net Access Code  $SAC_{AS_{new}}$  to the AS. It then computes:  $m_{GS2} = T_{GS2} || TUA_{GS1} || SAC_{GS1} || k_{ccGS2} || P_{GS2} || TUA_{AS} || SAC_{AS_{new}}$ . A random number  $N_2$  is generated, and  $m_{GS2}$  is encrypted using the AEAD encryption algorithm with  $k_{GS1,GS2}$ :  $(c_{m_{GS2}}, t_{m_{GS2}}) \leftarrow AEAD_{Encrypt}(k_{GS1,GS2}, N_2, m_{GS2})$ . Finally, GS2 sends  $N_2$ ,  $c_{m_{GS2}}$  and  $t_{m_{GS2}}$  to GS1.
- **Step 3: GS1 sends the essential information of GS2 along with the token to the AS.** GS1 receives  $N_2$ ,  $c_{m_{GS2}}$  and  $t_{m_{GS2}}$  and decrypts the message using the shared DCH session key  $k_{GS1,GS2}$ :  $m_{GS2} \leftarrow AEAD_{Decrypt}(k_{GS1,GS2}, N_2, c_{m_{GS2}}, t_{m_{GS2}})$ . First, GS1 verifies the timestamp  $T_{GS2}$ . GS1 then measures the current timestamp  $T_{token}$  and incorporates this timestamp along with the authentication result of GS1 for AS, encrypted with the LDAN public key, into the token distributed to the AS. The token is defined as:  $Token = T_{token} || verifyresult_{pubLDAN}$ . GS1 then computes:  $m_{GS1} = SAC_{AS} || token || SAC_{AS_{new}} || P_{GS2} || k_{CCGS2}$ . A random number  $N$  is generated based on the

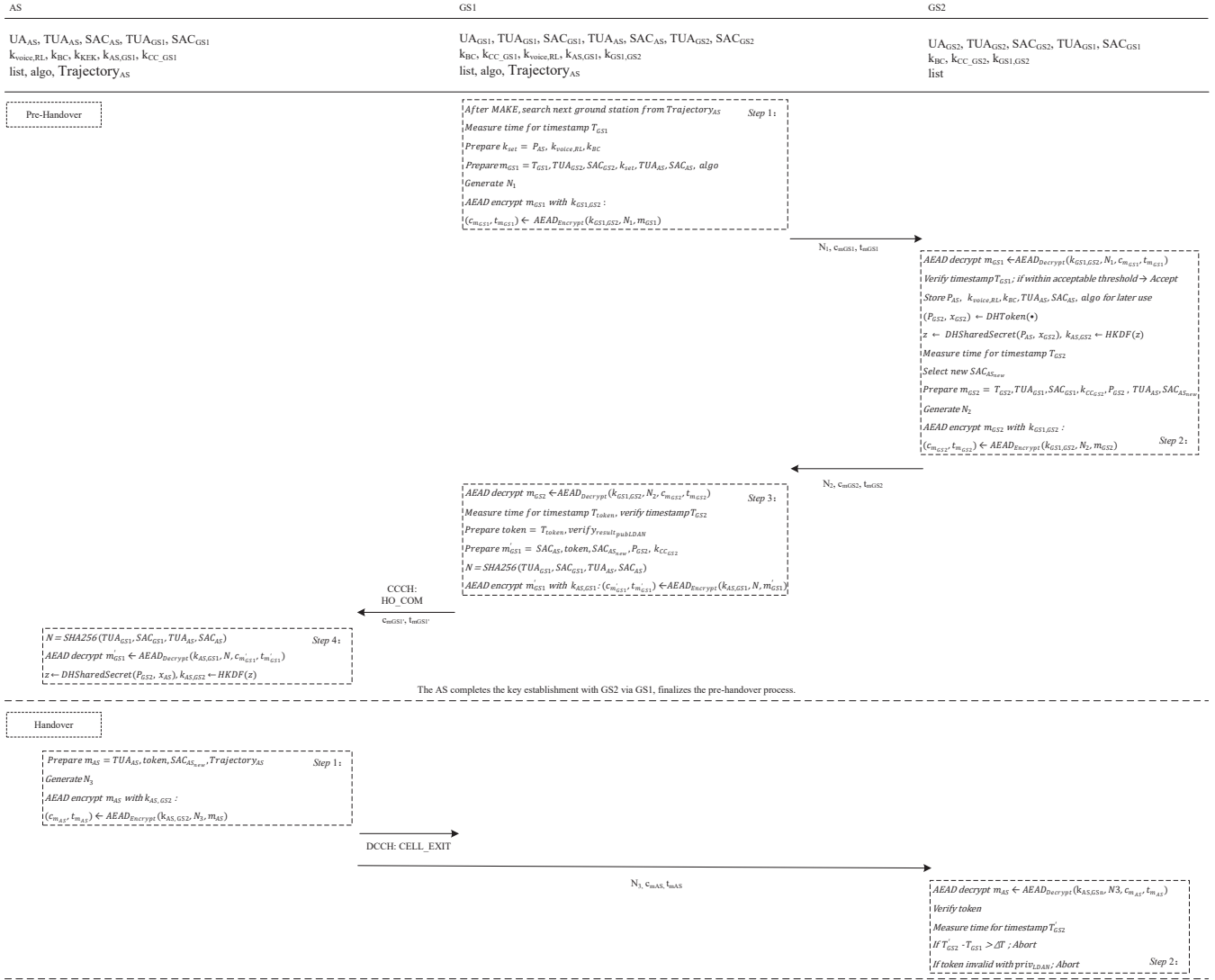


Fig. 2. Secure and low-latency handover protocol for LDACS.

identity information of GS1 and AS as follows:  $N = SHA256(TUA_{GS1} || SAC_{GS1} || TUA_{AS} || SAC_{AS})$ . The message  $m'_{GS1}$  is then encrypted using the key  $k_{AS,GS1}$  through the AEAD encryption algorithm:  $(c_{m'_{GS1}}, t_{m'_{GS1}}) \leftarrow AEAD_{Encrypt}(k_{AS,GS1}, N, m'_{GS1})$ . Finally,  $c_{m'_{GS1}}$  and  $t_{m'_{GS1}}$  are transmitted to AS via the CCCH in the handover command.

- **Step 4: The AS completes the key establishment process with GS2.** GS1 receives  $c_{m'_{GS1}}$  and  $t_{m'_{GS1}}$ . A random number  $N$  is generated by AS using the identity information of GS1 and AS as follows:  $N = SHA256(TUA_{GS1} || SAC_{GS1} || TUA_{AS} || SAC_{AS})$ . AS decrypts the received data using the shared DCH session key  $k_{AS,GS1}$ :  $m'_{GS1} \leftarrow AEAD_{Decrypt}(k_{AS,GS1}, N, c_{m'_{GS1}}, t_{m'_{GS1}})$ . AS then computes the shared secret key  $z$  using  $P_{GS2}$  and its private key  $x_{AS}$ :  $z \leftarrow DHSharedSecret(P_{GS2}, x_{AS})$ .

The shared secret  $z$  is used to derive the session key  $k_{AS,GS2}$  through the HKDF key derivation algorithm:  $k_{AS,GS2} \leftarrow HKDF(z)$ . At this stage, AS successfully completes key establishment with GS2 through GS1.

## B. Handover

- **Step 1: The AS presents the token to access GS2.** When the aircraft approaches a region where the GS1 signal weakens, it initiates a handover to GS2 following its trajectory. The AS computes the message:  $m_{AS} = TUA_{AS} || token || SAC_{AS,new} || Trajectory_{AS}$ . A random number  $N_3$  is generated, and the message  $m_{AS}$  is encrypted using the DCH session key established with GS2:  $(c_{m_{AS}}, t_{m_{AS}}) \leftarrow AEAD_{Encrypt}(k_{AS,GS2}, N_3, m_{AS})$ . Subsequently,  $N_3$ ,  $c_{m_{AS}}$ , and  $t_{m_{AS}}$  are transmitted to GS2.
- **Step 2: GS2 verifies the token and completes the authentication of the AS.** GS1 receives  $N_3$ ,

$c_{m_{AS}}$  and  $t_{m_{AS}}$ . It decrypts the received message using the shared DCH session key  $k_{AS,GS2}$ :  $m_{AS} \leftarrow AEAD_{Decrypt}(k_{AS,GS2}, N_3, c_{m_{AS}}, t_{m_{AS}})$ . First, GS2 verifies the timestamp in the token. If the timestamp falls within the acceptable threshold, the process continues; otherwise, it terminates immediately. Subsequently, the GS2 decrypts the GS1-to-AS authentication result embedded in the token using the LDAN private key. If the verification fails, the process is halted immediately.

At this point, the AS successfully completes the secure handover between ground stations. GS2 finalizes the authentication of the AS and completes key establishment, while ensuring the anonymity of station identities within the system.

#### IV. SECURITY ANALYSIS

This section presents a comprehensive security analysis of the proposed protocol, encompassing both informal and formal methodologies. The informal analysis employs a systematic approach articulated in natural language to evaluate security properties qualitatively. Conversely, the formal analysis utilizes rigorously defined semantic frameworks and strict syntactic structures to quantitatively validate the protocol's robustness against potential threats.

##### A. Informal Security Analysis: Logical Analysis

a) *Mutual Authentication*: The handover protocol proposed in this study ensures mutual authentication between the AS and GS2. Firstly, GS2 establishes a secure connection with GS1, and its identity is authenticated by GS1. Consequently, GS1 facilitates the AS's authentication of GS2. During the pre-handover stage, GS1 encrypts the authentication result of the AS using the LDAN public key and includes it in the token distributed to the AS, effectively preventing malicious tampering with the token's contents. Subsequently, during the AS-initiated handover process, GS2 validates the token using the LDAN private key to retrieve GS1's authentication result for the AS. This mechanism completes the mutual authentication between the AS and GS2.

b) *Ensuring Anonymity*: The proposed protocol effectively safeguards the true identities of both airborne and ground platforms from disclosure. During the AS's access to the ground network, the AuC assigns a TUA to each entity and promptly updates the ground access network and adjacent ground stations regarding TUA changes. The AuC maintains a secure mapping table linking TUAs to actual UAs, which is updated during each access process, thereby mitigating the risk of TUA leakage. Throughout the handover process, all identity-related information is transmitted exclusively through anonymized TUAs, preventing the potential theft or exposure of real UAs and mitigating risks such as malicious tracking. Therefore, the proposed handover protocol ensures robust anonymity for all entities, significantly enhancing security.

c) *Establishing Session Key Establishment*: Key establishment, including the derivation of  $k_{AS,GS2}$  and other keys, is facilitated through a key exchange algorithm and the HKDF key derivation algorithm, with GS1 acting as an intermediary

TABLE II  
TAMARIN VERIFICATION RESULTS

| Lemma                    | Scope  | Result |
|--------------------------|--------|--------|
| Session Exists           | Exists | ✓      |
| 2 Sessions Exist         | Exists | ✓      |
| Mutual Authentication    | All    | ✓      |
| Perfect Forward Secrecy  | All    | ✓      |
| Session Keys Consistency | All    | ✓      |
| Session Uniqueness       | All    | ✓      |

<sup>a</sup> ✓: Goal achieved ✗: Goal unachieved.

between the AS and GS2. The key derivation process incorporates the identity information of both the AS and GS2, thereby enhancing the confidentiality of the derived keys.

d) *Message Integrity and Confidentiality*: The proposed protocol utilizes the AEAD encryption algorithm, where the sender's computation process employs the shared DCH session key between stations as the encryption and decryption key, generating the tag  $t$  and ciphertext  $c$ . During the decryption process, the receiver uses the AEAD algorithm to verify the integrity of the information through the tag and the confidentiality of the information through the ciphertext. If either requirement is not met, the decryption process fails.

e) *Ensuring Message Freshness*: During each information transmission, the current timestamp is included in both the message and the *token* issued by GS1 to the AS. Upon receiving the signaling, the receiver immediately verifies the timestamp. If it falls within an acceptable time difference threshold, the process continues; if it exceeds the allowable range, the operation is immediately terminated. This mechanism helps prevent threats such as replay attacks from malicious entities, while enhancing the security of the handover process between AS and ground stations.

##### B. Formal Security Analysis: Tool Validation

Tamarin Prover is a tool for symbolic modeling and analysis of security protocols. It takes a protocol model as input, specifying the actions of protocol participants (such as the initiator, responder, and trusted server), the capabilities of attackers, and the security properties required by the protocol. Users can then use Tamarin to prove the security of the protocol, specifically whether the protocol satisfies its specified security properties when multiple instances of protocol participants and the behavior of attackers occur concurrently. The verification results of the proposed LDACS handover protocol using Tamarin Prover are shown in Table II.

As can be seen from the Table II, we tested the successful tracking in both single and multiple sessions, implementation of mutual authentication between AS and GS2, allowing attackers to retrieve data from previous sessions, session keys consistency and uniqueness. These Tamarin tests were conducted on a Ubuntu 18.04 machine with 16GB RAM and an Intel(R) Core(TM) i5-13400 processor.

Based on this, it can be concluded that the LDACS anonymity and trajectory-based handover protocol we designed successfully achieves mutual authentication, key estab-

TABLE III  
COMPARISON OF SECURITY FEATURES

| Goals                 | [5] | Proposed |
|-----------------------|-----|----------|
| Mutual Authentication | ✓   | ✓        |
| Anonymity             | ✗   | ✓        |
| Key Exchange          | ✓   | ✓        |
| Message Integrity     | ✓   | ✓        |
| Confidentiality       | ✓   | ✓        |
| Message Freshness     | ✓   | ✓        |
| Security Analysis     | ✓   | ✓        |

<sup>a</sup> ✓: Goal achieved ✗: Goal unachieved.

lishment and privacy protection between entities, ensuring the integrity and confidentiality of the communication system's information.

## V. PERFORMANCE AND COMPARATIVE ANALYSIS

In this section, we will discuss performance comparison, including the security of the LDACS secure handover protocol solution we propose, along with the associated latency and communication cost.

### A. Security Features

When comparing the proposed protocol with traditional LDACS handover protocols in terms of security, as shown in Table III, previous handover protocol, while able to accomplish mutual authentication and key establishment between airborne and ground platforms, ensuring the integrity and confidentiality of messages, fails to achieve identity anonymity for all stations within the system, making them vulnerable to malicious tracking and interference. The proposed protocol, while ensuring identity authentication and data encryption, utilizes temporally updated temporary identity codes between stations to protect identity information during signaling transmission, thereby achieving system anonymity. This enhancement improves the LDACS secure handover protocol mechanism, enabling it to better resist external attacks such as spoofing and replay. Moreover, our protocol effectively validates the timeliness of received information through timestamp verification, ensuring its validity within the specified time frame.

### B. Handover Latency

Latency is crucial for the handover process between aircraft and LDACS ground stations. To minimize the security handover latency, this paper divides the handover process into two stages: pre-handover and handover. During the pre-handover stage, key establishment between AS and GS2 is completed in advance because GS2 is predictively pre-selected following the aircraft's trajectory planned. And simultaneously, AS receives the authentication token issued by GS1. When AS is about to leave AS1 and access GS2 to initiate the handover stage, GS2 only needs to refer to the token content to verify AS's identity. As a result, the air-ground latency during the handover process only requires a single RL signaling communication from AS to GS2. Thus, the handover latency in the proposed protocol consists of three parts: the AS processing time, the RL latency, and the GS2 processing time.

TABLE IV  
COMPARISON OF HANDOVER LATENCY

| Protocol | Security Level [17] | Latency |
|----------|---------------------|---------|
| [5]      | 1 pre-q             | 328 ms  |
|          | 2 pre-q             | 328 ms  |
| Proposed | 1 pre-q             | 153 ms  |
|          | 2 pre-q             | 153 ms  |

<sup>a</sup> The latency of proposed protocol includes: Computation for Security at AS (<0.2ms),  $L_{RL}(t)$  (152.5ms) and Computation for Security at GS2 (<0.3ms).

To calculate the LDACS FL air-ground latency, this paper utilizes a latency simulation model specifically designed for LDACS [16]. The FL latency in the model is calculated as follows

$$L_{FL}(t) = m_{FL}(t) + (1 + \delta_{RX}(1 + n)) \times d_{MF}, \quad (1)$$

where,  $m_{FL}(t) = U(1, 60)$  represents the time remaining until the start of the next control channel frame,  $\delta_{RX}$  is a binary indicator flag signaling the need for retransmission,  $d_{MF} = 60ms$  denotes the duration of a Multi Frame, and  $n = \#AS/32$  is derived from the length of the reverse link's medium access cycle, as observed from the forward link.

The RL latency is calculated as follows

$$L_{RL}(t) = m_{RL}(t) + (2 + \delta_{RX}(N + 3)) \times d_{MF}, \quad (2)$$

where,  $m_{RL}(t) = \#AS/32 \times d_{MF} + U(1, 60)$  represents the time remaining until the commencement of the next dedicated control slot, with  $\delta_{RX}$  indicating whether retransmission is required. The term  $d_{MF} = 60ms$  denotes the duration of a Multi Frame, and  $N = (\#AS/32 - 3) \bmod \#AS/32$  corresponds to the reverse link medium access cycle length.

We model LDACS have maximum 32 AS per DC slot per MF, and  $\#AS$  is the amount of AS per LDACS cell. We model  $\delta_{RX}$ , which signifies the need for retransmission, as a binary stochastic process influenced by the packet error rate. This error rate is derived from the bit error rate (BER) and the packet length  $l$ . The probability that a packet is transmitted without errors is given by:  $P(\text{no error in packet}) = (1 - BER)^l$ . Conversely, the probability that a packet contains errors and thus requires retransmission is:  $P(\text{error in packet}) = 1 - (1 - BER)^l$ . These probabilities determine the value of  $\delta_{RX}$ . When  $\delta_{RX} = 1$ , it indicates that an error has occurred in the packet, necessitating retransmission. Retransmitting the packet ensures the reliability and integrity of data communication, which is critical in aeronautical communication systems where data loss or errors could lead to significant operational consequences.

As mentioned earlier, the latency of the LDACS secure handover protocol we designed consists of three parts: computation for security at AS, RL latency, and computation for security at GS2, as follows

$$L_{HO} = AS_{comp} + L_{RL}(t) + GS2_{comp}. \quad (3)$$

When considering the handover of a single aircraft between ground stations, running our protocol in the OMNet++ 6.0

TABLE V  
COMPARISON OF COMMUNICATION COST

| Protocol | Security Level | Pre-Handover | Handover |
|----------|----------------|--------------|----------|
| [5]      | 1 pre-q        | /            | 2115 bit |
|          | 2 pre-q        | /            | 3331 bit |
|          | 1 pre-q        | 2777 bit     | 1096 bit |
|          | 2 pre-q        | 3769 bit     | 1128 bit |

simulation, where the BER is  $10^{-6}$ , yields the mathematic expectation of latency results as shown in Table IV.

As shown in Table IV, the communication latency induced by LDACS RL is significantly higher than the signaling and verification processes between AS and GS2. Therefore, the handover delay can be simplified as follows

$$L_{HO} = L_{RL}(t) + constant. \quad (4)$$

It can be observed that the proposed trajectory-based pre-handover protocol significantly reduces the latency consumed during the AS preparation phase compared to traditional LDACS security handover protocols, i.e. the handover process requires only one single air-ground transmission. Specifically, the latency has been reduced from the original 328 ms [5] to approximately 153 ms, greatly improving the LDACS security handover latency performance.

### C. Communication Cost

According to the cryptographic primitive sizes for security types specified in the latest LDACS A/G standards, a comparison is made between the data overhead of the proposed LDACS security handover protocol and that of traditional protocols, as shown in Table V. We assume that the token is 320 bit in size, and the AS trajectory data transmitted includes the latitude and longitude information of three stations: the access station and its two adjacent stations, totaling 384 bit. In the pre-handover stage, the data overhead of the proposed protocol is 2777 bit and 3769 bit at different security levels, and in the handover stage, the data overhead is 1096 bit and 1128 bit at different security levels, resulting in a total data overhead of 3873 bit and 4897 bit. Although the total communication cost is higher than that of traditional protocol, the number of air-ground interactions and the data overhead during the handover process have been significantly reduced, minimizing the waste of spectrum resources and greatly improving the LDACS ground station handover efficiency.

## VI. CONCLUSIONS

In this paper, we propose a secure and low-latency handover protocol for LDACS ground stations, utilizing temporary UA updates, the distribution of token in the pre-handover stage, and token verification during the handover process. This approach effectively prevents the transmission of the real UA, while simultaneously reducing the latency of LDACS handover, thus conserving valuable aeronautical communication resources. We conducted a comprehensive security analysis, and the results demonstrate that the desired security attributes

have been successfully achieved. Performance analysis of the protocol reveals significant reductions in both the latency and communication cost during the LDACS handover process, offering valuable insights for the future development of LDACS security.

## ACKNOWLEDGMENT

The authors extend their sincere gratitude to the colleagues at the State Key Laboratory of CNS/ATM for their valuable insights and constructive feedback, which greatly contributed to the advancement of this research. This work was supported by the National Key Research and Development Program of China under Grant 2023YFB3907001.

## REFERENCES

- [1] Zhang J. Aeronautical Mobile Communication: The Evolution from Narrowband to Broadband[J]. Engineering, 2021, 7(4): 431-434.
- [2] Mäurer N, Guggemos T, Ewert T, et al. Security in digital aeronautical communications a comprehensive gap analysis[J]. International Journal of Critical Infrastructure Protection, 2022, 38: 100549.
- [3] Yu H, Wang S, Zhu Y, et al. L-DACS Mobility Management in ATN/IPS Network: Design, Prototyping, and Evaluation[C]//2023 IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC). IEEE, 2023: 1-7.
- [4] Mäurer N, Bilzhaus A. A cybersecurity architecture for the L-band digital aeronautical communications system (LDACS)[C]//2018 IEEE/AIAA 37th Digital Avionics Systems Conference (DASC). IEEE, 2018: 1-10.
- [5] Maurer N, Graupl T, Schmitt C, et al. A secure ground handover protocol for LDACS[C]//Proceedings of International Workshop on ATM/CNS 2022 International Workshop on ATM/CNS. Electronic Navigation Research Institute, 2022: 1-8.
- [6] Khan S, Gaba G S, Gurtov A, et al. Post quantum secure handover mechanism for next generation aviation communication networks[J]. IEEE Transactions on Green Communications and Networking, 2024.
- [7] Yang Y, Xue W, Sun J, et al. PKT-SIN: A Secure Communication Protocol for Space Information Networks with Periodic k-Time Anonymous Authentication[J]. IEEE Transactions on Information Forensics and Security, 2024.
- [8] Ma R, Cao J, Feng D, et al. FTGPHA: Fixed-trajectory group pre-handover authentication mechanism for mobile relays in 5G high-speed rail networks[J]. IEEE transactions on vehicular technology, 2019, 69(2): 2126-2140.
- [9] Zhou J, Dan Z, Wang Y, et al. Impact of DME Interference on LDACS Channel Estimation in En-route Phase[C]//2023 IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC). IEEE, 2023: 1-6.
- [10] ICAO, Finalization of LDACS Draft SARPs - Working Paper WP05 including Appendix, Tech. rep., International Civil Aviation Organization (ICAO).
- [11] Mäurer N, Graeupl T, Schmitt C. L-band digital aeronautical communications system (LDACS)[J]. RFC 9372, 2023.
- [12] Khan S, Gaba G S, Gurtov A, et al. Enhancing Cybersecurity for LDACS: a Secure and Lightweight Mutual Authentication and Key Agreement Protocol[C]//2023 IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC). IEEE, 2023: 1-10.
- [13] Morioka K, Kohmura A, Yonemoto N, et al. Rapid Prototyping and International Validation Activity for the L-Band Digital Aeronautical Communications System (LDACS)[J]. IEEE Open Journal of the Communications Society, 2023.
- [14] Gräupl T, Rihacek C, Haindl B, et al. LDACS A/G specification SESAR2020[J]. Tech. Rep. PJ14-02-01 D3. 3.030, 2019.
- [15] Liu P, Bai T, Dan Z, et al. A Lightweight and Anonymous MAKE for LDACS[C]//2024 16th International Conference on Wireless Communications and Signal Processing (WCSP). IEEE, 2024: 684-690.
- [16] Mäurer N, Gräupl T, Gentsch C, et al. Comparing different Diffie-Hellman key exchange flavors for LDACS[C]//2020 AIAA/IEEE 39th Digital Avionics Systems Conference (DASC). IEEE, 2020: 1-10.
- [17] Mäurer N, Gräupl T, Gentsch C, et al. A secure cell-attachment procedure of LDACS[C]//2021 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW). IEEE, 2021: 113-122.